Machine Learning

CS-871
Lecture-1

Instructor

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Graduate course for students of

Artificial Intelligence
Electrical Engineering
Cyber Security & Data Sciences

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Large Language Models
Renewable Energy
Energy Policy

Total Publications	37
Total Impact Factor	**121**
IF Journal Publications (W-Category)	27
International Conferences	09
Scopus Journal Publications	02
National Conferences	02
Peer Reviews of Journals	70+
Conference Reviews	15+
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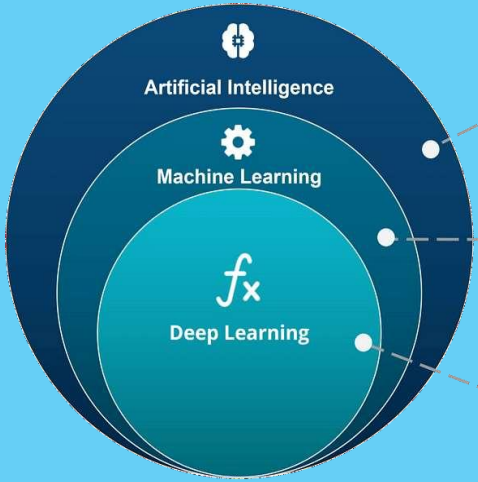


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AI & Machine Learning

AI and Subsets



ARTIFICIAL INTELLIGENCE
A technique which enables machines to mimic human behaviour

MACHINE LEARNING
Subset of AI technique which use statistical methods to enable machines to improve with experience

DEEP LEARNING
Subset of ML which make the computation of multi-layer neural network feasible

What is Machine Learning?

“Learning is any process by which a system improves performance from experience.”

- Herbert Simon

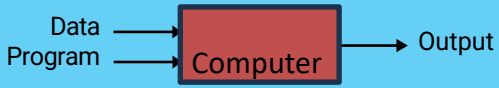
Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

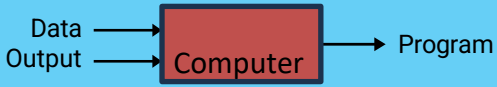
- improve their performance *P*
- at some tasks *T*
- with experience *E*.

A well-defined learning task is given by <*P*, *T*, *E*>.

Traditional Programming



Machine Learning



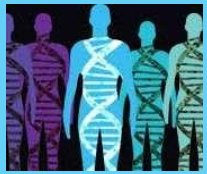
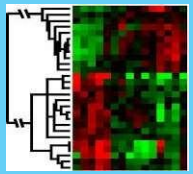
Slide credit: Pedro Domingos

When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



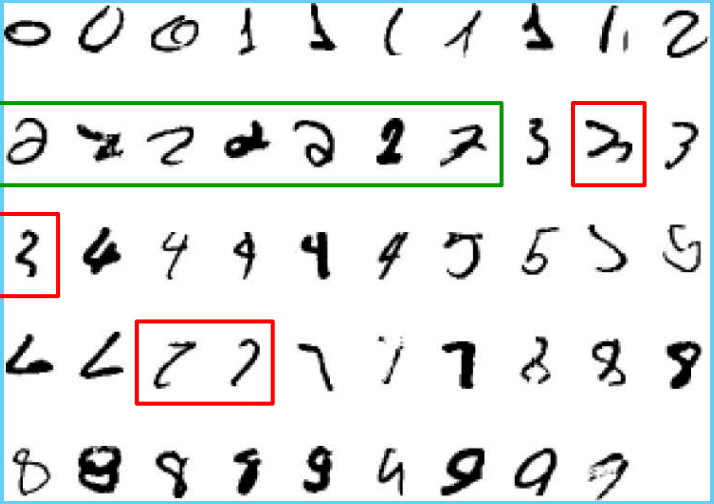



Learning isn't always useful:

- There is no need to "learn" to calculate payroll

Based on slide by E. Alpaydin

A classic example of a task that requires machine learning: it is very hard to say what makes a 2



Slide credit: Geoffrey Hinton

Some more examples of tasks that are best solved by using a learning algorithm

- **Recognizing patterns:**
  - Facial identities or facial expressions
  - Handwritten or spoken words
  - Medical images
- **Generating patterns:**
  - Generating images or motion sequences
- **Recognizing anomalies:**
  - Unusual credit card transactions
  - Unusual patterns of sensor readings in a nuclear power plant
- **Prediction:**
  - Future stock prices or currency exchange rates

Slide credit: Geoffrey Hinton

Defining the Learning Task

Improve on task T, with respect to
performance metric P, based on experience E

T: Playing checkers
P: Percentage of games won against an arbitrary opponent
E: Playing practice games against itself

T: Recognizing hand-written words
P: Percentage of words correctly classified
E: Database of human-labeled images of handwritten words

T: Driving on four-lane highways using vision sensors
P: Average distance traveled before a human-judged error
E: A sequence of images and steering commands recorded while observing a human driver.

T: Categorize email messages as spam or legitimate.
P: Percentage of email messages correctly classified.
E: Database of emails, some with human-given labels

Slide credit: Ray Mooney

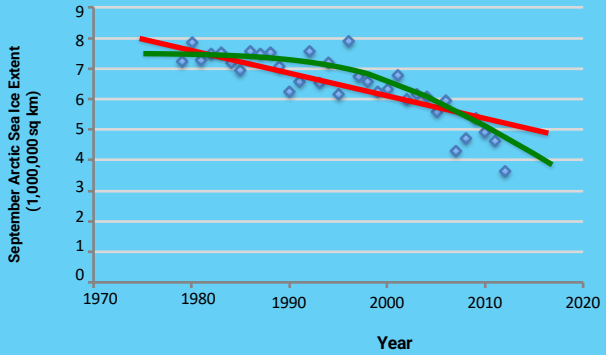
Types of Learning

- **Supervised (inductive) learning**
  - Given: training data + desired outputs (labels)
- **Unsupervised learning**
  - Given: training data (without desired outputs)
- **Semi- supervised learning**
  - Given: training data + a few desired outputs
- **Reinforcement learning**
  - Rewards from sequence of actions
- **Association Analysis**
  - Based on pattern mining

Based on slide by Pedro Domingos

Supervised Learning: Regression

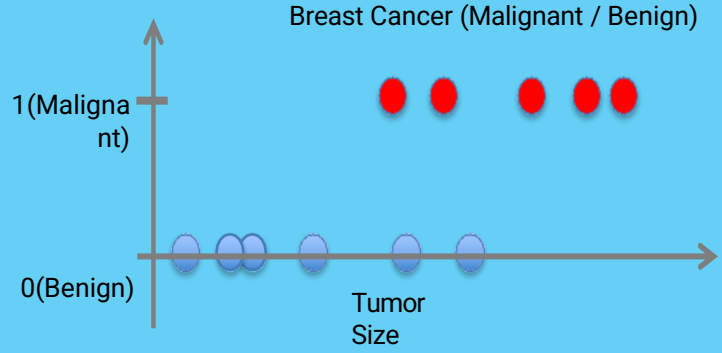
- **Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$**
- Learn a function $f(x)$ to predict y given x
  - y is real-valued == regression



Data from G. Witt. Journal of Statistics Education, Volume 21, Number 1 (2013)

Supervised Learning: Classification

- **Given (x_1, y_1), (x_2, y_2), ..., (x_n, y_n)**
- Learn a function $f(x)$ to predict y given x
  - y is categorical == classification

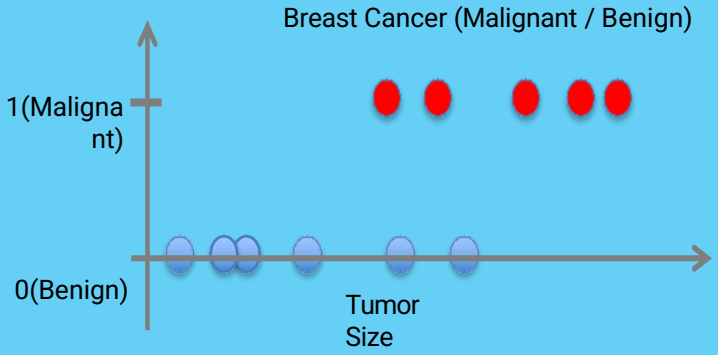


Based on example by Andrew Ng

Supervised Learning: Classification

- **Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$**

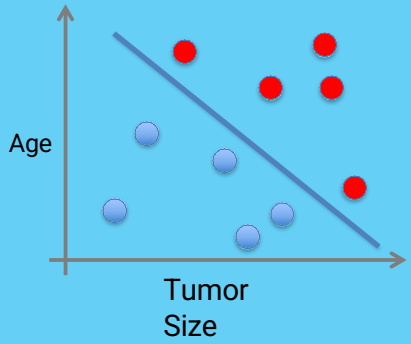
- Learn a function $f(x)$ to predict y given x
  - y is categorical == classification



Based on example by Andrew Ng

Supervised Learning

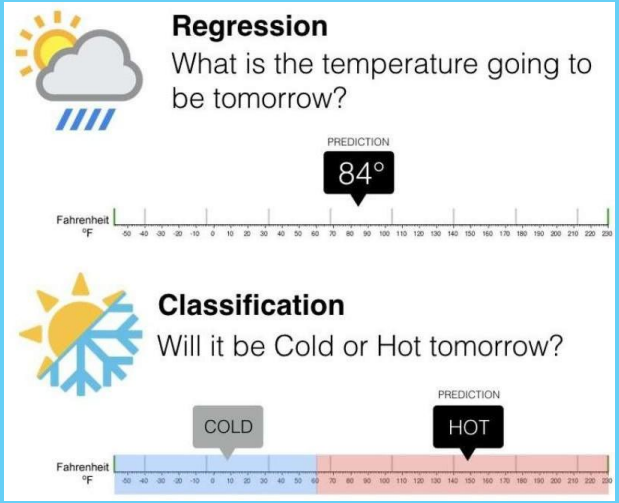
- *x* can be multi-dimensional
  - Each dimension corresponds to an attribute



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

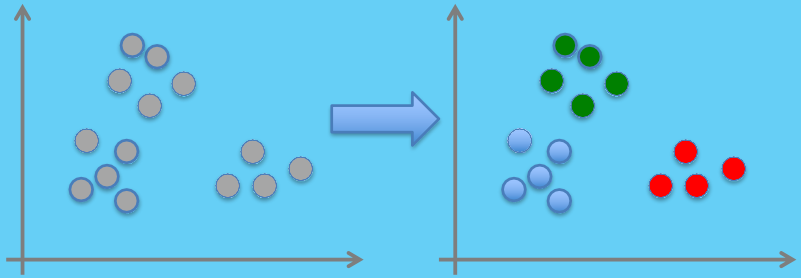
Based on example by Andrew Ng

Regression vs Classification



Unsupervised Learning

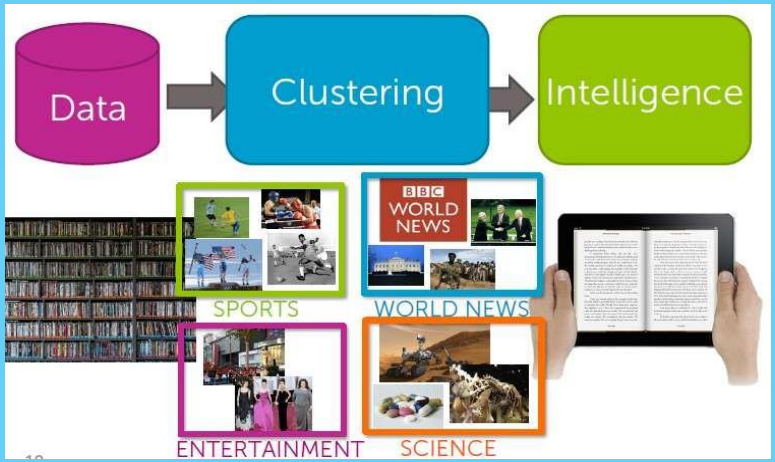
- Given x_1, x_2, ..., x_n(without labels)
- Output hidden structure behind the x's
  - E.g., clustering



Based on example by Andrew Ng

Unsupervised Learning

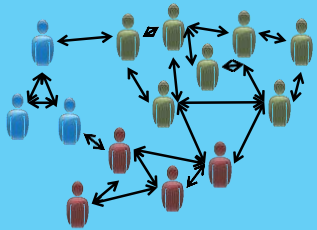
Document Retrieval



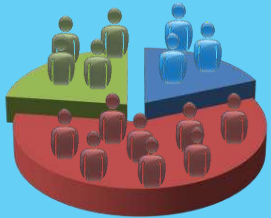
Unsupervised Learning



Organize computing clusters



Social



Market segmentation



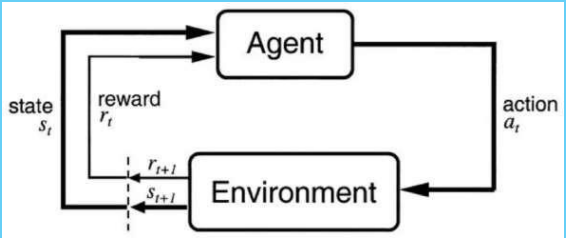
Astronomical data analysis

Slide credit: Andrew Ng

Reinforcement Learning

- Given a sequence of states and actions with (delayed) rewards, output a policy
  - Policy is a mapping from states to actions that tells you what to do in a given state
- Examples:
  - Credit assignment problem
  - Game playing
  - Robot in a maze
  - Balance a pole on your hand

The Agent Environment Interface



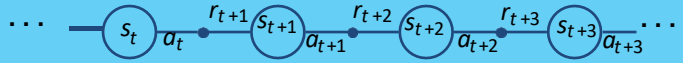
Agent and environment interact at discrete time steps : $t = 0, 1, 2, \mathrm{K}$

Agent observes state at step t: $s_t \in S$

produces action at step t : $a_t \in A(s_t)$

gets resulting reward : $R_{t+1} \in \Re$

resulting next state : s_{t+1}



Slide credit: Sutton & Barto

Reinforcement Learning



https://www.youtube.com/watch?v=4cgWya-wjgY

Learning Associations

Basket analysis:

$P(Y \mid X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

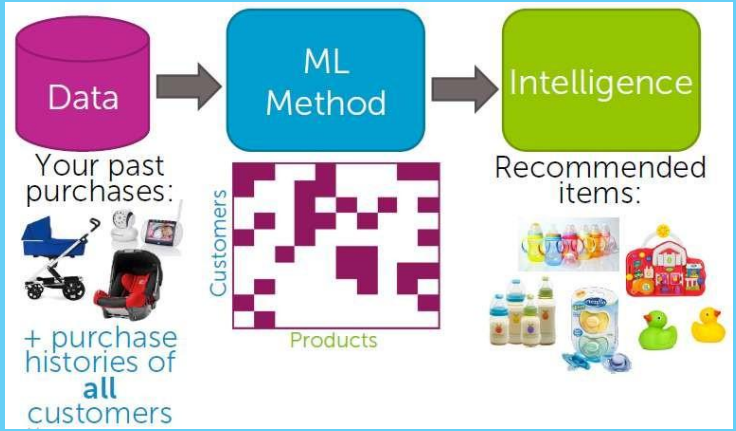
Example: P (bread |cold drink) = 0.7

Market-Basket transactions

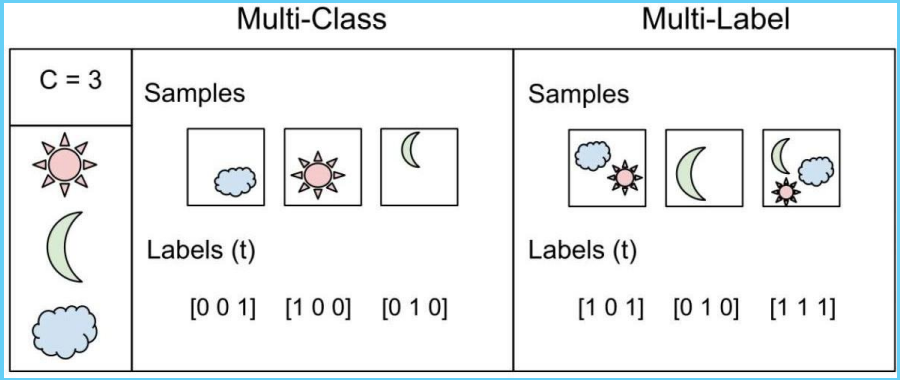
TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Cold Drink, Eggs
3	Milk, Diaper, Cold Drink
4	Bread, Milk, Diaper, Cold Drink
5	Bread, Milk, Diaper, Water

Learning Associations

Recommender Systems



Multi-Class vs Multi-Label



A brief history of machine learning

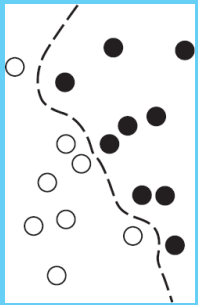
- ***Probabilistic Modeling***
  - The application of the principles of statistics to data analysis
  - **Naive Bayes** algorithm
    - Naive Bayes is a type of machine-learning classifier based on applying Bayes' theorem while assuming that the features in the input data are all independent (a strong, or "naive" assumption)
  - A closely related classification model is the ***logistic regression***, which can be considered as the "hello world" of modern machine learning

A brief history of machine learning

- ***Kernel methods***
  - A group of classification algorithms
  - Support Vector Machine (SVM) (best-known algorithm)
    - Aims at solving classification problems by finding good ***decision boundaries*** between two sets of points belonging to two different categories

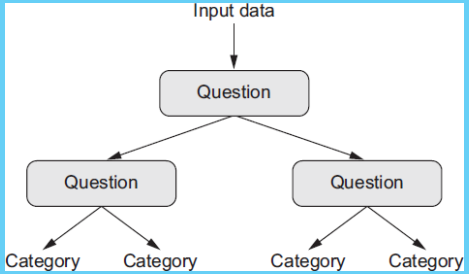
A decision boundary can be thought of as a line or surface separating your training data into two spaces corresponding to two categories

To classify new data points, you just need to check which side of the decision boundary they fall on



A brief history of machine learning

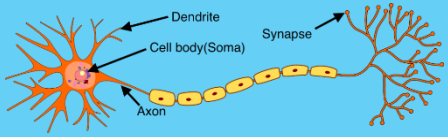
- ***Decision trees, random forests, and gradient-boosting machines***
  - *Decision trees* are flowchart-like structures that let you classify input data points or predict output values given inputs



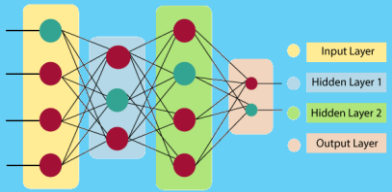
A decision tree: the parameters that are learned are the questions about the data. A question could be, for instance, “Is coefficient 2 in the data greater than 3.5?”

A brief history of machine learning

- ***Neural Networks***



- **Neurons:** computational building block for the brain



- **(Artificial)Neurons:** computational building block for the "neural network"

Key Difference:

- **Parameters:** Human brains have ~10,000,000 times synapses than artificial neural networks.
- **Topology:** Human brains have no "layers". **Async:** The human brain works asynchronously, ANNs work synchronously.
- Learning algorithm: ANNs use gradient descent for learning. We don't know what the human brain uses
- Power consumption: Biological neural networks use very little power compared to artificial networks
- Stages: Biological networks usually never stop learning. ANNs first train then test.

A brief history of machine learning

- ***Neural Networks***

  - Core ideas of neural networks were investigated as early as the 1950s

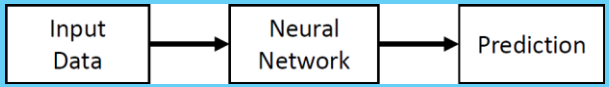
  - For a long time, the missing piece was an efficient way to train large neural networks

  - This changed in the mid-1980s when multiple people independently rediscovered the **Backpropagation algorithm**— a way to train chains of parametric operations using **gradient- descent optimization**—and started applying it to neural networks

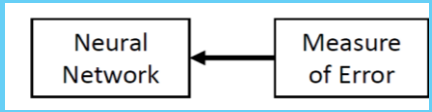
A brief history of machine learning

- ***Neural Networks***

Forward Pass:



Backward Pass (aka Backpropagation):

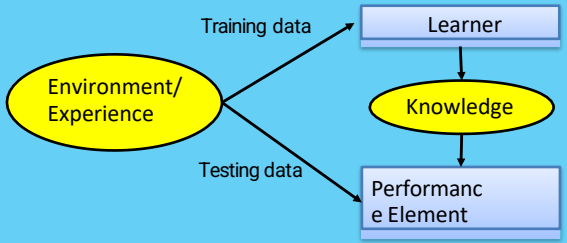


Adjust to Reduce Error

Framing a Learning Problem

Designing a Learning System

- Choose the training experience
- Choose exactly what is to be learned
  - i.e. the ***target function***
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience



Based on slide by Ray Mooney

Training vs. Test Distribution

- We generally assume that the training and test examples are independently drawn from the same overall distribution of data
  - We call this "i.i.d" which stands for "independent and identically distributed"
- If examples are not independent, requires

  collective classification

- If the test distribution is different, it requires
  transfer learning

Slide credit: Ray Mooney

ML in a Nutshell

- Tens of thousands of machine learning algorithms
  - Hundreds of new every year
- Every ML algorithm has three components:
  - **Representation**
  - **Optimization**
  - **Evaluation**

Slide credit: Pedro Domingos

Various Function Representations

- **Numerical functions**
  - Linear regression
  - Neural networks
  - Support vector machines
- **Symbolic functions**
  - Decision trees
  - Rules in propositional logic
  - Rules in first-order predicate logic
- **Instance-based functions**
  - Nearest-neighbor
  - Case-based
- **Probabilistic Graphical Models**
  - Naïve Bayes
  - Bayesian networks
  - Hidden-Markov Models (HMMs)
  - Probabilistic Context-Free Grammar (PCFGs)
  - Markov networks

Slide credit: Ray Mooney

Various Search/Optimization Algorithms

- **Gradient descent**
  - Perceptron
  - Backpropagation
- **Dynamic Programming**
  - HMM Learning
  - PCFG Learning
- **Divide and Conquer**
  - Decision tree induction
  - Rule learning
- **Evolutionary Computation**
  - Genetic Algorithms (GAs)
  - Genetic Programming (GP)
  - Neuro-evolution

Slide credit: Ray Mooney

Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- etc.

Slide credit: Pedro Domingos

ML in Practice



- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge

Based on a slide by Pedro Domingos

Lessons Learned about Learning

- Learning can be viewed as using direct or indirect experience to approximate a chosen target function.
- Function approximation can be viewed as a search through a space of hypotheses (representations of functions) for one that best fits a set of training data.
- Different learning methods assume different hypothesis spaces (representation languages) and/or employ different search techniques.

Slide credit: Ray Mooney

Machine learning overview

Overview

* we describe here the general ideas and methods used in machine learning, at a high level
* we will go over all of these topics later, in much more detail
* don’t worry if some of this seems abstract at this point, or there are terms you don’t know yet

Artificial intelligence approaches

- we would like computers to perform complicated tasks, *e.g.*, medical diagnosis
- distinguish two approaches
  - *knowledge-based*: a computer program whose logic encodes a large number of properties of the world, usually developed by a team of experts over many years
  - *machine learning*: extract information directly from historical data and extrapolate to make predictions
- this class is about machine learning

Machine learning tasks

generic tasks:

* build a *model* from some *data*
  * choose how to map raw data to feature vectors
  * choose a model form
  * choose parameter values in the model
* *test* or *validate* the model
  * evaluate the model on unseen data to assess its performance

‘model’ can mean several things, depending on context

Machine learning model taxonomy: supervised vs. unsupervised

* supervised learning models *predict something, given some other things*
  * called a *prediction model*
  * called *regression* when we predict a real scalar or vector value
  * called *classification* when we predict a value from a finite set such as (TRUE/ FALSE)
  * called *forecasting* when you predict a future value, given current and past values

* unsupervised learning models just create a model of the data
  * called a *data model*

* the lines between these can be blurred

Machine learning model taxonomy: point vs. probability

* a *point estimate* predicts a single value
* a *probability estimate* predicts a distribution of values
* a *confidence band estimate* predicts a confidence band or interval of values
* a *generative model* generates samples from an estimated distribution

* the lines between these can be blurred

Examples

what kind of models would each of these tasks use?

* predict tomorrow's rainfall, given the date and the last 10 days of rainfall data
* determine from a photo of a face if the user is who she claims to be
* estimate the probability of 10 possible diagnoses, given some patient data, test results
* cluster customers into 22 different groups with similar buying habits
* estimate the risk (probability) of an auto accident at a location given the hour and weather
* build an *anomaly detector*, that rates how suspicious some new data is
* build a *simulator* that generates fake new data that 'looks like' the given data
* build a *recommendation engine* that suggests products a customer might be interested in

Performance metrics

- we judge performance of a model on some data using a *metric* such as
  - mean-square or RMS prediction error (for regression)
  - error rate (for classification)
  - log likelihood (for probabilistic models)
- examples:
  - our predictor predicts tomorrow's maximum temperature with an RMS error of 1%
  - our classifier predicts the topic of a newspaper article (from a set of 50 choices) with an error rate of 5%

Training and validation

* our goal is to develop a model that performs well for *new, unseen data*
* standard practice is to divide the given data set into two parts
  * a *training* data set, used to choose or train the model
  * a *test* or *validation* data set, used to evaluate tentative models
* we can look at the performance metrics on the *training* and *test* data sets
* if the model performs well on the training set, but poorly on the test set, it is *overfit*
  (and probably useless in practice)
* if the model performs well on the test set, it's likely going to perform well on new unseen data

Learning a model

a common method of choosing a model:

* choose the *model structure* or *form* or *type*
* the model contains a number of *model parameters*
* choose a *loss function* that rates how badly the model performs on a single data point or example
* choose the parameter value by minimizing an average loss over the training data

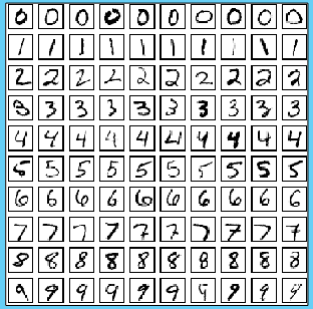
this general scheme, called *empirical risk minimization*, is used to fit a wide variety of models

Examples

Example: diagnosis

- goal is to predict if a patient has a disease, based on whether or not she exhibits 10 symptoms
- historical data consists of a large number of *patient records*
- each record contains
  - 10 Booleans, specifying the presence or absence of the 10 symptoms
  - a Boolean specifying whether that patient had the disease
- machine learning algorithm observes these data, produces a *predictor*
- predictor takes as input 10 Booleans, returns a single Boolean prediction
- this is a *classifier*, since we are predicting an outcome that takes only two values
- we will judge model by its error rate on a separate test set of data, not used to develop the model
- a *probabilistic model* returns a probability that the patient has the disease, not just a Boolean

Example: digit recognition



- 8-bit grayscale images of handwritten digits, 28×28 pixels
- goal is to guess the digit (*i.e.*, $0, \ldots, 9$) from the image
- 60,000 training images, 10,000 test images
- preprocessed by antialiasing, scaling and centering
- data originally by NIST, modified by Le Cun, 1998

Data

* Kaggle: datasets and competitions
* ImageNet dataset: 14m images
* Street view house numbers: 600,000 digit images
* Waymo open dataset: self-driving car data
* many other large datasets

Software

* PyTorch, Keras, TensorFlow, JAX, and many others, . . .
* Scikit-Learn, Spark/MLlib
* Flux, in Julia
* generic packages in R, Matlab, Python, Java, Julia . . .

Course outline

The major parts of the course

1. *regression* — to predict one or more real values
2. *classification* — to predict one of a finite number of possible outcomes
3. *probabilistic supervised learning* — to predict a distribution of outcomes
4. *unsupervised learning* — to develop a data model
5. *optimization* — to fit or choose parameters in all of the models above